DEPARTMENT-WISE SECURE NETWORK SEGMENTATION

PROBLEM STATEMENT:

Many enterprises use a flat network architecture with little or no segmentation between departments. Users and devices across Finance, HR, IT, Sales, and other units can communicate freely without strict access controls. This increases the risk of unauthorized access and attacker lateral movement within the network. Malware or ransomware incidents can spread quickly, causing widespread impact. Lack of segmentation makes it difficult to enforce the principle of least privilege and meet security compliance standards. Implementing proper network segmentation and controlled inter-department communication reduces risks and protects sensitive data.

SYSTEM ARCHITECTURE:

a) Core Design Approach

- Network Segmentation using VLANs / Subnets
  - Separate VLANs for Finance, HR, IT, Sales, Management, and Guest networks.
- Layered Security Model (Defense-in-Depth)
  - Perimeter Firewall
  - Internal Segmentation Firewall (ISFW)
  - Access Control Lists (ACLs)
- Zero Trust Architecture (ZTA)
  - Verify every user/device before granting access.
- Network Access Control (NAC)
  - Device authentication before joining the network.

- DMZ (Demilitarized Zone)
    - Public-facing servers isolated from internal network.
- SIEM & Monitoring Layer
    - Centralized logging and real-time threat detection.

b) Logical Architecture Layers

1. Perimeter Layer
    - Next-Generation Firewall (NGFW)
    - IDS/IPS
2. Core Network Layer
    - Layer 3 Switches with VLAN segmentation
    - Internal firewalls between departments
3. Access Layer
    - NAC-enabled switches
    - Endpoint security agents
4. Security & Monitoring Layer
    - SIEM
    - Security Operations Center (SOC)

TOOLS AND TECHNOLOGIES:

Network Segmentation

- VLANs (IEEE 802.1Q)
- Subnetting (IPv4/IPv6)
- VRF (Virtual Routing and Forwarding)

Firewalls & Security

- Next-Generation Firewalls (e.g., Palo Alto Networks, Fortinet)
- Internal Segmentation Firewalls (ISFW)
- IDS/IPS systems (e.g., Cisco solutions)

Access Control

- Network Access Control (NAC)
    - Example: Cisco Systems Identity Services Engine (ISE)
- Role-Based Access Control (RBAC)
- Multi-Factor Authentication (MFA)

Monitoring & Detection

- SIEM (e.g., Splunk)
- Endpoint Detection & Response (EDR)
- Security Orchestration and Automation (SOAR)

Additional Technologies

- VPN with secure tunneling
- Micro-segmentation (Software-Defined Networking)
- Zero Trust Network Access (ZTNA)

PROPOSED SOLUTION

1. Implement VLAN-Based Department Segmentation
    - Create separate VLANs for each department.
    - Restrict inter-VLAN communication using firewalls.
2. Deploy Internal Firewalls
    - Apply granular access control policies.
    - Allow only necessary services between departments.

3. Adopt Zero Trust Model
    - Authenticate and authorize every user and device.
    - Apply least privilege policies.
4. Enable Network Access Control (NAC)
    - Ensure only compliant and authorized devices connect.
5. Deploy Continuous Monitoring
    - Use SIEM for real-time log analysis.
    - Implement IDS/IPS for intrusion detection.
6. Establish DMZ for Public Services
    - Isolate web servers, email gateways, and external-facing systems.
7. Regular Security Audits & Policy Review
    - Conduct vulnerability assessments and penetration testing.

EXPECTED OUTCOMES:

- Reduced risk of lateral movement by attackers.
- Limited impact of malware/ransomware outbreaks.
- Improved protection of sensitive departmental data (Finance, HR).
- Strong enforcement of the Principle of Least Privilege.
- Improved regulatory compliance (ISO 27001, PCI-DSS, HIPAA, etc.).
- Better visibility into network traffic and faster incident response.
- Enhanced overall enterprise cybersecurity posture.